



COMPONENTS

e-Paper displays

Pervasive Displays' Aurora black-and-white e-paper, or E-ink, electrophoretic displays now available are chip-on-glass (COG) front plane laminate (FPL) displays that make use of waveform technology to produce sharp images in a thin design. Only updating the display's image requires power, allowing total power consumption to be kept to an absolute minimum. This makes them ideal for applications where the content of the display changes irregularly, such as digital signs, retail spaces, office automation, building navigation, reusable badges, and healthcare. The full range of displays from 3.7cm to 31cm is available through the company's distribution partner DigiKey.

Pervasive Displays, Taiwan
www.pervasivedisplays.com

Wi-Fi integrated front end

Mouser Electronics is now stocking the QPF7219 integrated front end module (iFEM) from Qorvo. Providing reliable, whole-home coverage when designed into Wi-Fi 6 (802.11ax) systems, it combines advanced bulk acoustic wave (BAW) filter technology with its unique edgeBoost capability to almost double Wi-Fi range and triple capacity to handle more devices. It integrates a 2.4GHz power amplifier optimised for a 5V supply voltage.

The BAW filter's edgeBoost capability maximises output power at channels close to the band edge and achieves excellent throughput across all Wi-Fi channels 1-11. The module includes a bypassable low-noise amplifier, regulator, and transmit-receive switch. The device offers 5GHz rejection for dual-band dual-concurrent operation, as well as integrated die-level filtering for second and third harmonics. It is ideal for wire-

less routers, access points, residential gateways, Wi-Fi customer premises equipment, and IoT applications.

Mouser Electronics
www.mouser.com

Voltage regulators

Infineon Technologies AG IR3887M, IR3888M and IR3889M devices in its new family of integrated-point-of-load voltage regulators feature fast constant-on-time (COT) engine. This product family is designed for today's server, base station and telecom (operating at 85°C ambient temperature) and storage applications requiring both high efficiency and high density. The IR3887M is the smallest 30A device on the market in a 4mm × 5mm footprint. The OptiMOSTM IPOL product family comes with an enhanced-stability COT engine with ceramic capacitors without the need for external compensation. Wide input voltage range of 4.3V to 17V, 2Vmin with 5V external bias, and precise output voltage (0.5 per cent accurate reference voltage) are supported. The voltage regulators offer high efficiency at light-load and thermally-compensated internal over-current protection with four selectable limits for a less expensive and smaller inductor. High switching frequency (up to 2MHz) for small PCB size and fewer bill-of-materials is claimed. The IR3888M and IR3889M come in 5mm x 6mm footprint in a PQFN package. The IR3888M handles continuous current up to 25A and is RoHS2 compliant.

Infineon Technologies India Pvt Ltd
www.infineon.com

AD converter technology

Renesas Electronics, in collaboration with Hitachi, has developed a technology that will enable continuous-time digital calibration of a delta-sigma ($\Delta\Sigma$) modulator and an analogue-to-digital (AD) converter circuit. This will make high-speed, high-precision $\Delta\Sigma$ AD converters

capable of withstanding rough conditions. To solve the difficulty in achieving higher precision, the technology digitally calibrates the transfer function of the $\Delta\Sigma$ modulator. High-speed, high-precision operation with a 15MHz signal bandwidth and 74.3dB dynamic range is achieved when operating at an oversampling frequency of 480MHz. By reducing the digital calibration circuit operating frequency to 120MHz, a low-power operation is also achieved, with 37mW power consumption (analogue:19mW, digital:18mW). In addition, the new, robust technology provides stable performance over a wide temperature range of -20 to 125 degrees Celsius.

Renesas Electronics Corp.
www.renesas.com

DC-DC buck converter

Texas Instruments has introduced 40A SWIFT DC-DC PMBus buck converter TPS546D24A that offers stacking of up to four ICs. It can deliver up to 160A of output current (4x current of competing power ICs) at a temperature of 85 degrees Celsius. The TPS546D24A has the highest efficiency of any 40A DC-DC converter, thus allowing engineers to reduce power loss by 1.5W in a high-performance data centre and enterprise computing, medical, wireless infrastructure and wired networking applications. The TPS546D24A offers a switching frequency of 1.5MHz for delivering 40A of current per IC while reducing inductance and capacitance by one-third as compared to similar converters. It has an output voltage error of less than one per cent.

Texas Instruments
www.ti.com

Industrial connectors

Amphenol ICC, in collaboration with Hirose Electric and Harting Technology, has released the ix Industrial series of connectors. IEC 61076-3-124 compliant and designed to meet the needs of next-generation rugged, high-speed industrial Ethernet requirements, the series is ideal for use in equipment for factory and process automation, machine-to-machine